

A/B testing rigor

Week 5, Lecture 1 · B2C to \$10k MRR: A 10-Week Growth Manual

Autumn 2026

What we will cover

- What an A/B test actually controls for
- Sample-size math: when you do NOT have enough traffic
- The painted-door test: the low-traffic alternative
- The peeking problem and why p-values lie
- What to test (and what not to) in weeks 1-10

Part 1: what an A/B test controls for

The only thing an A/B test proves

- An A/B test proves that variant B caused a difference in metric Y (or didn't)
- "Caused" requires: random assignment, simultaneous exposure, one change at a time
- Without randomization, you have correlation, not causation
- Kohavi, Tang & Xu (2020): "Controlled experiments are the gold standard for establishing causal relationships."

What randomization actually does

- Randomization balances all user characteristics across control and variant
- Users who signed up on Tuesday, users on mobile, users with free plans: equally distributed
- Without randomization: "users who saw the new page" might just be newer users with different behavior
- The feature flag is how you enforce randomization at scale

One change, one metric

- Change only one thing per test: the UI, the copy, the price, the flow
- Measure one primary metric: not "engagement and retention and revenue"
- Secondary metrics are fine to watch, but call the winner on one number only
- Multiple changes + multiple metrics = you cannot tell what moved what

Part 2: sample-size math

The sample-size heuristic

For a binary metric (conversion rate, activation rate):

```
n per variant = 16 × variance / MDE2
```

```
variance = baseline × (1 - baseline)
```

```
MDE = minimum detectable effect (absolute percentage points)
```

This gives you 80% power at a 5% false-positive rate. Two variants means $2 \times n$ total users.

A worked example

- Activation rate: 20% (baseline = 0.20)
- You want to detect a 4pp improvement (MDE = 0.04)
- Variance = $0.20 \times 0.80 = 0.160$
- $n \text{ per variant} = 16 \times 0.160 / (0.04^2) = 2.56 / 0.0016 = 1,600$
- Total users needed: 3,200

If you get 400 uniques per week: **8 weeks to run this test.** That's too long.

What "too long" means

- Tests longer than 4 weeks are usually not worth running
- Novelty effects, seasonality, and user mix shift change the result
- If $n / \text{weekly traffic} > 4$ weeks: you do not have enough traffic to A/B test this change
- The right move: painted-door test, qualitative interview, or just ship with a kill switch

Part 3: the painted-door test

What a painted-door test is

- Show users a button, feature, or option that does not yet exist
- Measure click-through rate as a proxy for demand
- After the click: "coming soon" page or waitlist sign-up
- Eric Ries (2011): the original lean startup validation technique

What a painted-door test is NOT

- It is not an A/B test: there is no control group
- It does not prove a feature will improve retention or revenue
- It proves that some users are interested enough to click
- Useful threshold: if fewer than 10% click, reconsider building it

The honest painted-door

- Always close the loop: after the click, tell users the feature is coming
- Never leave users in a dead end with no explanation
- Collect their email or ask one question: "what would you use this for?"
- The qualitative follow-up is often more valuable than the click rate

Part 4: peeking and p-hacking

The peeking problem

- If you check results every day and stop when $p < 0.05$, your false-positive rate is not 5%
- PostHog team (2023): Airbnb discovered this at scale; peeking inflates false positives substantially
- The standard 5% threshold assumes you look once, at the end
- Every additional peek is effectively a new test

What p-hacking looks like in practice

- Checking results daily and stopping when things "look good"
- Switching the primary metric after seeing the data
- Running many tests and only reporting the winners
- Kohavi et al. call this "statistical significance theater"

The fix: pre-register before you start

- Write down before touching code: primary metric, MDE, run duration, decision rule
- Do not look at variant-level breakdowns until the run period ends
- Set a calendar reminder for the end date
- Sequential testing (Wald, or tools like Statsig's sequential test) is the rigorous alternative if you need early stopping

Part 5: what to test in weeks 1-10

Good A/B test candidates

- Onboarding flow: a single step or a single piece of copy
- Activation: the first screen a new user sees after sign-up
- Paywall: hard vs. soft, trial length, price point display
- These touch the activation-rate and conversion-rate metrics you can already measure

Bad A/B test candidates (for now)

- Features that affect $<5\%$ of your users: not enough exposure
- Changes to the core product loop when retention data is thin
- Anything that requires >4 weeks to reach sample size
- Price changes if you have fewer than 100 paying customers: the cohort is too small

Most experiments fail. Bing sees only a 10-20% success rate on their tests, but each failure provides learning opportunities.

<footer>Ian Vanagas, PostHog, "10 things we've learned about A/B testing for startups," 2023</footer>

Take-aways

1. An A/B test proves causation only if you have random assignment and one change at a time.
2. Compute n per variant = $16 \times \text{variance} / \text{MDE}^2$ before you set up any test. If run time exceeds 4 weeks, use a painted-door test instead.
3. Never read results before your pre-registered end date. Peeking inflates false positives.
4. Most tests fail. That is not a problem: a failed test is a valid learning, not wasted time.

What's next

- **Reading:** Week 5 reading covers sample-size math, phased rollouts, and the mid-course check
- **Lecture 2 (this week):** CI/CD for one, feature flags, kill switches, rollback runbooks
- **Section (this week):** Design one A/B test: sample-size arithmetic on your own product
- **Friday:** 60-minute mid-course pivot/persevere check (details in the reading)